

INDUSTRY PROBLEM SOLVING TASK

1. Autonomous Vehicle Vision System Selection

An automobile company is developing an autonomous vehicle detection system that must operate under the following conditions:

- If accuracy $< 90\%$ $\rightarrow$ system fails
- If latency > 70 ms $\rightarrow$ unacceptable
- The vehicle supports moderate computational resources

Methods available:

- Viola-Jones: 82% accuracy, 30 ms latency, low computation
- YOLO: 93% accuracy, 60 ms latency, moderate computation
- Deep Learning: 97% accuracy, 90 ms latency, high computation

Task:

Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, latency, and computational requirements.

2. Smart Factory Product Inspection Selection

A manufacturing company wants to deploy a real-time computer vision system to detect defective products on a production line. The system must operate under the following conditions:

- If accuracy $< 95\%$ $\rightarrow$ defects may be missed
- If processing time > 50 ms $\rightarrow$ production is affected
- Hardware supports moderate computation

Methods available:

- Traditional Image Analysis: 91% accuracy, 25 ms, low computation
- YOLO: 96% accuracy, 45 ms, moderate computation
- Deep Learning: 98% accuracy, 75 ms, high computation

Task:

Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, processing time, and computational requirements.

1. Autonomous Vehicle Vision System Selection

Introduction

An autonomous vehicle needs a computer vision system that can detect objects such as vehicles, pedestrians, traffic signs, and obstacles quickly and accurately. Since the vehicle makes decisions in real time, both **accuracy and latency** are very important. The available computational resources are also limited to a moderate level. Therefore, the selected method must satisfy **all three conditions**.

Given Constraints

The autonomous vehicle has the following requirements:

Accuracy must be at least 90%.

If accuracy is below 90%, the system fails because it may incorrectly detect or miss important objects.

Latency must not exceed 70 ms.

If latency is more than 70 ms, the system becomes too slow for real-time vehicle operation.

Computational requirement must be moderate or low.

The vehicle supports moderate computational resources, so a method requiring very high computation is not suitable.

Comparison of Methods

Method	Accuracy	Latency	Computation	Accuracy Condition	Latency Condition	Computation Condition
Viola-Jones	82%	30 ms	Low	✗Fail	✓Pass	✓Pass
YOLO	93%	60 ms	Moderate	✓Pass	✓Pass	✓Pass
Deep Learning	97%	90 ms	High	✓Pass	✗Fail	✗Fail

1. Viola-Jones

Viola-Jones provides **82% accuracy**, which is below the required 90%. Although its latency is only **30 ms**, making it very fast, and it requires only **low computation**, its low accuracy makes it unsuitable.

In an autonomous vehicle, an accuracy of 82% could result in important objects not being detected correctly. For example, if a pedestrian or another vehicle is missed, it could create a serious safety problem.

Therefore:

Accuracy: **82% → Fail**

Latency: **30 ms → Pass**

Computation: **Low → Pass**

Since all conditions must be satisfied, Viola-Jones **cannot be selected**.

2. YOLO

YOLO provides **93% accuracy**, which is higher than the required 90%. Its latency is **60 ms**, which is also below the maximum allowed 70 ms. It requires **moderate computation**, which matches the computational resources available in the vehicle.

YOLO therefore provides a good balance between accuracy, speed, and computational requirements.

Therefore:

- Accuracy: **93% → Pass**
- Latency: **60 ms → Pass**
- Computation: **Moderate → Pass**
- YOLO satisfies **all the required conditions**.

3. Deep Learning

The Deep Learning method provides the highest accuracy of **97%**, which is excellent for object detection. However, it has a latency of **90 ms**, which is greater than the maximum allowed 70 ms.

It also requires **high computational resources**, while the vehicle only supports moderate computation. Therefore, even though its accuracy is better than YOLO, it cannot satisfy the complete system requirements.

Therefore:

- Accuracy: 97% → Pass
- Latency: 90 ms → Fail
- Computation: High → Fail
- Hence, Deep Learning cannot be selected.

Final Decision

YOLO is the most suitable method for the autonomous vehicle vision system.

The main reason is that YOLO is the **only method that satisfies all three constraints simultaneously**. It provides **93% accuracy**, which is above the required 90%; **60 ms latency**,

which is below the 70 ms limit; and **moderate computational requirements**, which are supported by the vehicle.

Although Deep Learning has better accuracy, its high latency and high computational requirement make it unsuitable. Viola-Jones is fast and computationally efficient, but its accuracy is too low.

Conclusion

Therefore, **YOLO should be selected** for the autonomous vehicle. It provides the best balance between **accuracy, speed, and computational efficiency**. For a real-time autonomous driving application, satisfying all constraints is more important than achieving the highest possible accuracy alone.

2. Smart Factory Product Inspection Selection

Introduction

A smart factory uses computer vision to automatically inspect products on a production line. The system must identify defective products accurately while processing products quickly. If the system is slow, production may be delayed. If the system has low accuracy, defective products may pass through the production line.

Therefore, the selected computer vision method must satisfy the requirements for **accuracy, processing time, and computational resources**.

Given Constraints

The smart factory has the following requirements:

Accuracy must be at least 95%.

- Accuracy below 95% may cause defective products to be missed.
- **Processing time must not exceed 50 ms.**

- A processing time above 50 ms can slow down the production line.
- **Computational requirement must be moderate or low.**
- The available hardware supports moderate computation.

Comparison of Methods

Method	Accuracy	Processing Time	Computation	Accuracy Condition	Time Condition	Computation Condition
Traditional Image Analysis	91%	25 ms	Low	✗Fail	✓Pass	✓Pass
YOLO	96%	45 ms	Moderate	✓Pass	✓Pass	✓Pass
Deep Learning	98%	75 ms	High	✓Pass	✗Fail	✗Fail

1. Traditional Image Analysis

Traditional Image Analysis has an accuracy of **91%**. Although it is fast, with a processing time of only **25 ms**, its accuracy does not meet the required 95%.

For a manufacturing company, missing defective products can cause quality problems and customer complaints. Therefore, the high speed of this method cannot compensate for its insufficient accuracy.

Therefore:

- Accuracy: **91% → Fail**
- Processing time: **25 ms → Pass**
- Computation: **Low → Pass**

Since the accuracy requirement is not satisfied, Traditional Image Analysis **cannot be selected**.

2. YOLO

YOLO provides **96% accuracy**, which is above the required 95%. It has a processing time of **45 ms**, which is below the maximum allowed 50 ms.

YOLO also requires **moderate computation**, which matches the hardware available in the smart factory.

Therefore:

- Accuracy: **96% → Pass**
- Processing time: **45 ms → Pass**

- Computation: **Moderate** → **Pass**
- YOLO satisfies **all three conditions**.

It is therefore capable of detecting defective products accurately while maintaining the required real-time processing speed.

3. Deep Learning

Deep Learning provides the highest accuracy of **98%**, which is above the required 95%. However, its processing time is **75 ms**, which exceeds the maximum allowed 50 ms.

The method also requires **high computational resources**, while the factory hardware supports only moderate computation.

Therefore:

- Accuracy: **98%** → **Pass**
- Processing time: **75 ms** → **Fail**
- Computation: **High** → **Fail**

Although Deep Learning has better accuracy, it cannot be selected because it violates the processing-time and computational-resource requirements.

Final Decision

YOLO is the most suitable method for the smart factory product inspection system.

YOLO provides **96% accuracy**, which is sufficient to detect defective products reliably. Its processing time of **45 ms** is within the 50 ms limit, allowing the production line to operate efficiently. It also requires only **moderate computation**, which is supported by the available hardware.

Traditional Image Analysis is faster but does not provide enough accuracy. Deep Learning provides higher accuracy but is too slow and computationally expensive.

Conclusion

Therefore, **YOLO should be selected** for the smart factory product inspection system. It provides the best balance between **accuracy, processing speed, and computational requirements**. Since it satisfies all the given constraints, it is the most practical choice for real-time defect detection on a production line.